

53 essential commands for OpenShift Virtualization & MTV

The commands I used to migrate a VMware VM to OpenShift Virtualization, grouped the way you need them.

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- 02 Cluster health & access

 - 03 Storage & Virtualization

 - 04 VM networking & MTV

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Cluster health & access

```
oc whoami --show-server
```

Which cluster am I talking to?

```
oc get clusterversion
```

Version and upgrade status

```
oc get co
```

Health of every platform component

```
oc get nodes -o wide
```

Nodes, IPs, OS image, runtime

```
oc adm top nodes
```

CPU and memory use per node

```
oc get mcp
```

Node OS config rollout (MachineConfigPools)

```
oc debug node/<node> -- chroot /host
```

The supported shell on an RHCOS node

```
oc get csv -A
```

Installed operators and their phase

```
oc get subscription -A
```

Which channel each operator follows

```
oc explain plan.spec.type
```

Field docs for the version you run

```
oc get users; oc get identity
```

Who has logged in, from which IdP

```
oc adm policy add-cluster-role-to-user cluster-admin <user>
```

Grant cluster admin

```
oc auth can-i create vm -n <ns> --as <user>
```

Test RBAC before the user does

Storage & Virtualization

```
oc get sc
```

StorageClasses and the default

```
oc get pvc,pv -A
```

Disks (claims) and volumes

```
oc get storageprofile <sc> -o yaml
```

Access mode CDI will use (want RWX)

```
oc get dv -A
```

DataVolume import progress

```
oc get hco -n openshift-cnv
```

Is Virtualization healthy?

```
oc get datasource -n openshift-virtualization-os-images
```

Golden boot images

```
oc get virtualmachineclusterinstancetype
```

VM sizes (instance types)

```
oc get vm,vmi -A
```

Defined VMs vs running VMs

```
virtctl start <vm> / virtctl stop <vm>
```

Power on / off

```
virtctl console <vm>
```

Serial console

```
virtctl migrate <vm>
```

Live migration: the new vMotion

```
oc get vmim -A
```

Live migrations in progress

```
oc get vmi <vm> -o wide
```

Which node runs this VM

```
oc patch vm <vm> --type merge -p '{"spec":{"runStrategy":"Always"}}'
```

Start a VM without virtctl

VM networking & MTV

```
oc get nncp
```

Node network policies (bridges, bonds)

```
oc get nnce
```

Did the policy apply on each node?

```
oc get nns <node> -o yaml
```

Live NIC state of a node

```
oc get net-attach-def -A
```

VM networks: the new port groups

```
oc get providers -n openshift-mtv
```

vCenter and cluster connections

```
oc describe provider <p> -n openshift-mtv
```

Why a provider is not Ready

```
oc get networkmap,storagemap -n openshift-mtv
```

Port group → NAD, datastore → SC

```
oc get plan -n openshift-mtv
```

Migration plans and status

```
oc create -f migration.yaml
```

Start a run (the Start button)

```
oc get migration -n openshift-mtv -w
```

Watch a run

```
oc describe plan <p> -n openshift-mtv
```

Pipeline steps, timings, errors

```
oc patch provider <p> -n openshift-mtv --type merge -p '{"spec":{"settings":{"vddkInitImage":"<img>"}}}'
```

Add VDDK: fast disk reads

```
oc logs <v2v-pod> -n <target-ns>
```

virt-v2v conversion log

```
oc logs deploy/forklift-controller -c inventory -n openshift-mtv
```

vCenter inventory errors

Troubleshooting & the bastion

```
oc get events -n <ns> --sort-by=.lastTimestamp
```

What just happened, newest last

```
oc get pods -A | grep -vE 'Running|Completed'
```

Only the unhealthy Pods

```
oc describe vmi <vm>
```

Why a VM will not schedule

```
oc adm must-gather
```

Support bundle for Red Hat

```
oc project default
```

Escape a context stuck on a deleted project

```
ip -br addr; ip route
```

IPs and routes on the bastion

```
ip route get <ip>
```

Which route the kernel really uses

```
dig +short api.<cluster>.<domain>
```

Do the cluster DNS records resolve?

```
timeout 4 bash -c "</dev/tcp/<esxi>/902"
```

Port test with no extra tools

```
sudo exportfs -v
```

What the NFS server exports

```
chronyc sources
```

Is time sync working? (^* good, ^? bad)

```
lsblk -f; df -h
```

Disks, filesystems, free space

The full field manual is free: 10 chapters, real terminal captures, GOTCHA boxes and EX316-style labs.

github.com/Willey2003/openshift-vmware-migration-sop